Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 19 · Part D: Digital Signal Processing Level 4

Track: Hypothesis Testing, Matched Filter, ROC, Whitening, Template Banks, False-Alarm Rate

Prerequisites: Problems 07, 13, 14 · **Estimated effort:** 14–20 hours

Matched Filters and Detection Theory: Deciding That Something Is There

A detection is a decision under uncertainty, and the optimal decision rule is known. Derive the matched filter from the Neyman–Pearson lemma, verify its performance by Monte Carlo, whiten coloured noise, search a template bank for a chirp, calibrate the false-alarm rate on noise-only data including the trials factor, and run the whole machine on open gravitational-wave data.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/19_matched_filters_and_detection/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

Radar, gravitational-wave interferometers, GPS receivers and gamma-ray burst monitors all ask the same question: is a known kind of signal hidden in this noisy record, and with what confidence? Statistical decision theory gives the answer. For a known waveform in Gaussian noise the optimal detector is a correlation with that waveform, the *matched filter*, and its performance depends on a single number, the signal-to-noise ratio. For unknown parameters (arrival time, mass, frequency) one maximises over a bank of templates, which brings a price: the more places you look, the more false alarms you get.

This problem develops the theory and then practises the discipline that makes a detection claim credible: calibrate the background on data that cannot contain the signal, count trials honestly, and compare with the theoretical optimum. It is the quantitative backbone of the transient-detection capstone (problem 25).

Theoretical Framework

1. Hypothesis Testing

Choose between H_0: $y = n$ and H_1: $y = s + n$ for a known s. The Neyman–Pearson lemma states that the test maximising the detection probability P_d at fixed false-alarm probability P_{fa} is the likelihood-ratio test. For white Gaussian noise of variance σ^2 it reduces to the correlator

$$T(y) = \sum_n y[n]\, s[n] \;\underset{H_0}{\overset{H_1}{\gtrless}}\; \gamma, \qquad P_d = \mathrm{Q}\big(\mathrm{Q}^{-1}(P_{fa}) - d\big), \qquad d^2 = \frac{\sum_n s[n]^2}{\sigma^2} = \frac{E_s}{\sigma^2}, \tag{1}$$

where d is the deflection coefficient. The receiver operating characteristic (ROC) is P_d against P_{fa}.

2. Matched Filter

The filter $h[n] = s[N-1-n]$ maximises the output SNR at the sampling instant $n = N-1$; its output is the cross-correlation of y with s and the peak SNR is $d^2 = 2E_s/N_0$ (continuous time, one-sided noise density N_0). For an unknown arrival time the output is searched over all lags; it is computed efficiently with FFTs.

3. Coloured Noise and Optimal SNR

For Gaussian noise with one-sided PSD $S_n(f)$ define the noise-weighted inner product $(a|b) = 4\,\mathrm{Re}\int_0^\infty \tilde{a}(f)\tilde{b}^*(f)/S_n(f)\,\mathrm{d}f$. The optimal SNR of a signal h is

$$\rho^2 = (h|h) = 4\int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\mathrm{d}f, \tag{2}$$

and the matched-filter SNR time series is $\rho(t) = (s|h_t)/\sqrt{(h|h)}$. Equivalently, whiten both data and template by $1/\sqrt{S_n}$ and apply the white-noise matched filter. In the absence of signal $\rho(t)$ is Gaussian with unit variance.

4. Template Banks and Match

When the signal depends on parameters $\boldsymbol{\theta}$ (here chirp mass), search over a bank $\{h_{\boldsymbol{\theta}_i}\}$. The *match* between normalised templates $\hat{h}_1, \hat{h}_2$ is $\mathcal{M} = \max_{t_0}(\hat{h}_1|\hat{h}_{2,t_0})$; a signal whose parameters fall between templates loses a fraction $1-\mathcal{M}$ of its SNR. Banks are spaced so that the minimal match is, say, 0.97.

5. False Alarms and Trials

With N_{trials} effectively independent tests per record, the false-alarm probability of the loudest is $P_{fa}^{\mathrm{tot}} \approx 1-(1-P_1)^{N_{\mathrm{trials}}} \approx N_{\mathrm{trials}}P_1$ with $P_1 = \mathrm{Q}(\rho_{\mathrm{thr}})$ per trial. The number of independent trials in time is about $T\,B_{\mathrm{eff}}$ (duration times effective bandwidth) times the number of templates. Real noise has non-Gaussian glitches, so thresholds should be calibrated empirically from noise-only data (time slides, off-source segments), not from the Gaussian tail alone.

6. Pulse Compression

A linear-FM pulse of duration T and bandwidth B has time–bandwidth product BT; its matched filter compresses it to width $\sim 1/B$ with SNR gain BT (in power) over a simple pulse of the same energy, and radar range scales as $R_{\max} \propto \mathrm{SNR}^{1/4}$.

Data and Simulation Specification

Pulse in white noise	Gaussian pulse of width 20 samples in $\sigma = 1$ white noise, scaled to $d = 1, 2, 3, 4, 5$; 10^5 trials per point.
Mismatched template	Template widths $10, 15, 20, 30, 40$ samples against a signal of width 20.
Chirp in white noise	Linear chirp, $N = 4096$ samples, injected at a random position, per-sample SNR chosen to give $\rho = 8$.
Chirp in coloured noise	`sp_common.synth.chirp_in_noise(fs=4096, duration=16, mchirp_msun=30, target_snr=8)` (and 12, 20); the generator returns the injected optimal SNR, the template, and `toy_ligo_psd`.
Background	225 independent noise-only records of 16 s (one hour in total) from `colored_noise` with the toy PSD.
Template bank	Chirp masses 15–45 $M_\odot$ with `sp_common.synth.newtonian_chirp`.
Real data (strongly recommended)	GW150914 (H1 and L1, 32 s at 4 kHz) from GWOSC; estimate the PSD from off-source data around the event, whiten, band-pass 20 Hz–300 Hz, and filter with a chirp template of $\mathcal{M}_c \approx 28\,\mathrm{M}_\odot$.

Your Task

1. **Detection theory by simulation (Stage 2).** For the pulse in white noise, compute P_d at $P_{fa} = 10^{-3}$ for $d = 1, \ldots, 5$ by Monte Carlo and compare with Eq. (1). Plot ROC curves at $d = 2, 3, 4$ with Wilson intervals and confirm the analytic curve lies within them.
2. **Matched versus mismatched and energy detectors.** Measure the SNR loss of the mismatched templates and express it in dB and as a P_d loss at $P_{fa} = 10^{-3}$. Compare with an energy detector (sum of squares in the right window). Show how much more signal energy the energy detector needs for the same P_d.
3. **FFT matched filter for a chirp.** Implement the matched filter for the chirp in white noise by FFT cross-correlation. Normalise so that the noise-only output has unit variance (check over 1000 realisations), find the peak time and compare the empirical peak-SNR distribution with $\rho \pm 1$.
4. **Coloured noise and whitening.** For `chirp_in_noise`, estimate the PSD with Welch's method from a signal-free stretch, whiten data and template and form $\rho(t)$. Compare the recovered peak SNR with the injected ρ for 200 noise realisations at each of $\rho = 8, 12, 20$, and with the SNR obtained when you skip the whitening. Compare the timing error of the peak with $\sigma_t \approx 1/(2\pi f_{\mathrm{rms}}\rho)$ for the signal's effective bandwidth f_{rms}.
5. **False-alarm calibration.** Run the matched filter over the one hour of noise-only records, build the distribution of the loudest ρ per record and determine the threshold giving one false alarm per hour and per day (extrapolate with a Gaussian-tail model and state its validity). Compare with the single-trial $\mathrm{Q}(\rho_{\mathrm{thr}})$ and with the trials factor you calculate from TB_{eff}.
6. **Template bank.** Compute the match between templates of different chirp mass, build a bank with minimal match 0.97 across 15–45 $M_\odot$, count its templates and measure the SNR loss for injections at random masses. Show how the bank size raises the threshold needed for the same false-alarm rate.
7. **Real data.** Run the full pipeline on GW150914: whiten, filter, produce $\rho(t)$, report the peak time and SNR in both detectors and the time difference between them, and compare with the published values. Explain why a Newtonian template recovers a substantially lower SNR than the published matched-filter SNR of about 24.

Self-check targets

For $P_{fa} = 10^{-3}$ the threshold is $\mathrm{Q}^{-1}(10^{-3}) = 3.09$; with $d = 4$ the detection probability is $P_d = \mathrm{Q}(3.09 - 4) = 0.82$. A matched-filter output on noise only is $\mathcal{N}(0, 1)$ to within sampling error. The injected `target_snr` equals Eq. (2) for the generator's PSD and $\rho(t)$ at the true time must scatter around it with unit standard deviation. A pulse-compression gain of $BT = 1000$ is 30 dB.

Aerospace Application Link

A pulsed radar detects a target at range R with SNR $\propto R^{-4}$. Using Eq. (1), compute the SNR needed for $P_d = 0.9$ at $P_{fa} = 10^{-6}$ and the factor by which the maximum range increases when a linear-FM pulse of time–bandwidth product $BT = 1000$ replaces a simple pulse of the same peak power and duration ($10^{30/40} \approx 5.6$ in range for 30 dB of gain). Then use the same machinery to state, for a GRB-like count excess searched on three timescales over a day, the SNR threshold you would adopt and the trials factor you used.

Deliverables

D1 **Detection-theory simulation:** the P_d table and ROC curves with analytic overlays.

D2 **Mismatch and energy detector:** the SNR loss table and the energy-detector comparison.

D3 **Chirp matched filter:** white-noise FFT implementation, unit-variance check and the coloured-noise whitened version with the recovered-SNR histogram.

D4 **Background calibration:** the one-hour noise-only run, threshold-versus-FAR curve and the trials-factor analysis.

D5 **Template bank and real data:** the bank construction, match matrix and the GW150914 result.

D6 **Memo (300 words):** "What does a 5-sigma detection mean, and how did we count trials?" Cite at least two external sources.

Bonus (a) Implement the χ^2 signal-consistency test of Allen and show it rejects a loud glitch. (b) Detect a coherent signal in two detectors with a time-shift window and estimate the false-alarm rate from time slides. (c) Derive and simulate the optimal detector for Poisson counts (log-likelihood-ratio weights $\ln(1 + s/b)$).

Learning Resources

Detection and Estimation Theory

- Kay, S.M. *Fundamentals of Statistical Signal Processing, Vol. II: Detection Theory* (Neyman–Pearson, matched filter, deflection coefficient).
- Van Trees, H.L. *Detection, Estimation, and Modulation Theory, Part I.*
- Richards, M.A. *Fundamentals of Radar Signal Processing* (matched filtering, pulse compression, detection).

Gravitational-Wave Data Analysis

- Allen, B., Anderson, W.G., Brady, P.R., Brown, D.A., Creighton, J.D.E. (2012). "FINDCHIRP: an algorithm for detection of gravitational waves from inspiraling compact binaries." Phys. Rev. D 85, 122006, arXiv:gr-qc/0509116.
- Abbott, B.P. et al. (2016). "GW150914: first results from the search for binary black hole coalescence with Advanced LIGO." Phys. Rev. D 93, 122003.
- GWOSC tutorials on whitening and matched filtering: `https://gwosc.org/tutorials/`; PyCBC (optional): `https://pycbc.org`

Tools

- SciPy `signal.correlate`, `signal.welch`, `stats.norm.isf`: `https://docs.scipy.org/doc/scipy/reference/`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.